

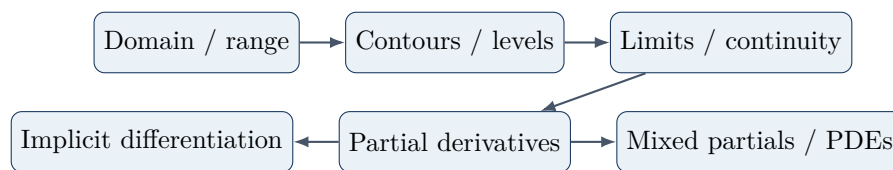
# Comprehensive Tutorial – Topics Covered So Far

AS1109: Calculus and Ordinary Differential Equations – Section H

Same tutorial for H1 (Wed 23 Sep, 4:00 pm) and H2 (Fri 25 Sep, 1:00 pm)

## Purpose of this 50-minute tutorial

This is a **single-session** revision tutorial. It is deliberately modelled on the *style and level* of the regular-admission Mid-Sem paper, but uses new questions. It revises almost all material actually taught to Section H so far: functions of several variables, domain/range, level curves and contours, limits, continuity, first/higher/mixed partial derivatives, the Mixed Derivative Theorem, implicit differentiation, and the Laplace/wave equations.



## Q1. Rapid concept check – answer in one or two lines each

[5 min]

- If  $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$ , must  $f(a,b)$  be defined?
- State the three conditions for continuity of  $f$  at  $(a,b)$ .
- If  $f_x(a,b)$  and  $f_y(a,b)$  both exist, must  $f$  be continuous at  $(a,b)$ ?
- State a sufficient condition that guarantees  $f_{xy}(a,b) = f_{yx}(a,b)$ .

## Q2. Domain, boundary, openness and range

[7 min]

Consider

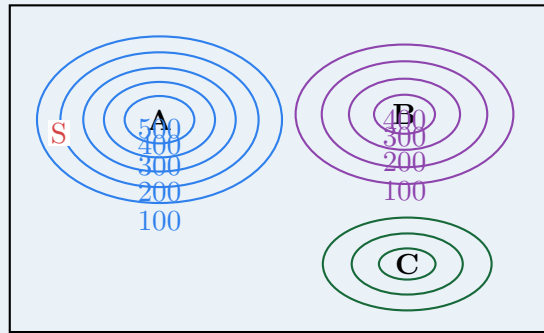
$$f(x,y) = \frac{1}{\sqrt{x^2 + y^2 - 16}}.$$

- Determine and sketch the domain.
- Find its boundary. Is the domain open, closed, both, or neither?
- Is the domain bounded or unbounded?
- Find the range of  $f$ .

## Q3. Reading a contour map

[5 min]

The diagram represents elevation in metres; the contour interval is 100 m.



- Which labelled peak is highest? Give a defensible elevation interval.
- Which hill has gentler overall slopes, A or B? Explain.
- Near which labelled location is the slope steepest: A, B, C or S? Explain.

## Q4. Limits and continuity – two Mid-Sem style techniques

[10 min]

- Determine whether

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$$

exists. Use two paths and state the logical conclusion clearly.

- For  $(x, y) \neq (0, 0)$  let

$$h(x, y) = \frac{\sin(5(x^2 + y^2))}{x^2 + y^2}.$$

Choose  $h(0, 0)$  so that  $h$  is continuous at the origin. Justify using polar coordinates.

## Q5. First, higher and mixed partials; Hessian

[10 min]

Let

$$g(x, y) = e^{xy} + \sin(x + y) + x^2y.$$

- Find  $g_x$  and  $g_y$ .
- Find  $g_{xx}, g_{xy}, g_{yx}, g_{yy}$  and write the Hessian matrix  $H_g(x, y)$ .
- Is the Hessian symmetric? Explain why the equality of the mixed partials is justified here.
- At a point, what is the geometric meaning of  $g_x$  and  $g_y$ ?

**Q6. Implicit differentiation****[7 min]**

The equation

$$xy + z^3y - 2xz = 0$$

defines  $z$  locally as a function of the independent variables  $x, y$ , and the required derivative exists.

- a) Derive a formula for  $z_x$ .
- b) Evaluate  $z_x$  at  $(x, y, z) = (1, 1, 1)$ .
- c) Derive a formula for  $z_y$  and evaluate it at  $(1, 1, 1)$ .

**Q7. Laplace and wave equations – recognition and verification****[6 min]**

- a) Verify that  $u(x, y) = e^{-x} \cos y$  satisfies Laplace's equation

$$u_{xx} + u_{yy} = 0.$$

Give one engineering/physical context in which Laplace's equation arises.

- b) Let  $v(x, t) = \cos(kx - \omega t)$ . Determine the relation between  $\omega$ ,  $k$  and  $c$  required for

$$v_{tt} = c^2 v_{xx}$$

to hold. What does  $c$  represent physically?

**If we finish early: 2-minute synthesis challenge**

Give an example of a function whose two first partial derivatives exist at  $(0, 0)$  but which is not continuous at  $(0, 0)$ . Explain the decisive path showing failure of continuity.

## One-look method map

| Problem type       | First move                                                                                           |
|--------------------|------------------------------------------------------------------------------------------------------|
| Domain/range       | Translate every formula restriction into inequalities/exclusions; then interpret the geometry.       |
| Contour map        | Height comes from labels; steepness comes from contour spacing.                                      |
| Two-variable limit | For non-existence, find two paths giving different limits; for existence, use a bound or polar form. |
| Continuity         | Check value, limit, and equality.                                                                    |
| Partial derivative | Hold the other independent variable fixed.                                                           |
| Mixed partials     | Differentiate in the stated order; equality needs hypotheses, not wishful thinking.                  |
| Implicit $z(x, y)$ | Differentiate the whole equation and remember $z$ depends on $x, y$ .                                |
| Laplace/wave PDE   | Compute exactly the second derivatives appearing in the equation and substitute.                     |